# MIS 637B- DATA ANALYTICS & MACHINE LEARNING

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Let's create a dataset of eight exam scores, ensuring that one score is an outlier.

Given Dataset:

Scores: 75, 82, 88, 90, 93, 95, 98, 40 (The score of 40 is an outlier)

**a.) Calculate the Mean and Median**

With the Outlier (40)

- Mean:

Mean= (75+82+88+90+93+95+98+40) / 8 = 661 / 8 = 82.625

- Median:

The middle two numbers (since there are 8 scores) are 88 and 90, so:

Median = (88+90) / 2 = 178 / 2 = 89

Without the Outlier (40)

- Mean:

Mean = (75+82+88+90+93+95+98) / 7 = 621 / 7 = 88.71

- Median:
  The middle value of the remaining seven scores is 90.

**b.) Effect of the Outlier**

- The mean drops significantly (from 88.71 to 82.625) when the outlier is included.
- The median remains relatively stable (from 90 to 89).

The mean is affected more because it takes into account every value in the dataset, including extreme values. The median, being a positional measure, is less influenced by a single outlier.

**c.) Verify Outlier Using IQR Method**

1. Find Q1 (First Quartile) and Q3 (Third Quartile):

- Q1: Median of the lower half (75, 82, 88, 90) → (82+88)/2 = 85
- Q3: Median of the upper half (93, 95, 98) → (93+95)/2 = 94

2. Compute Interquartile Range (IQR):

   IQR=Q3−Q1=94−85=9

3. Determine Outlier Boundaries:

- Lower Bound: $Q1 - 1.5 \times IQR = 85 - (1.5 \times 9) = 85 - 13.5 = 71.5$
- Upper Bound: $Q3 + 1.5 \times IQR = 94 + (1.5 \times 9) = 94 + 13.5 = 107.5$

4. Identify Outliers:

- Any value below 71.5 or above 107.5 is an outlier.
- The score 40 is below 71.5, so it is indeed an outlier.